

Kis és Középvállalati Hálózati Infrastruktúra Implementációs Dokumentáció

Cisco Packet Tracer Szimuláció

Készítette: Balogh Dávid

Dátum: 2026. 1. 13

Verzió: 1.0

Technológiák: Cisco IOS, OSPF, VLAN, ACL, IPsec VPN, DHCP

1. BEVEZETÉS

Ez a dokumentum egy középvállalati hálózati környezet tervezését és konfigurálását írja le. A projekt célja egy biztonságos, skálázható és redundáns infrastruktúra létrehozása volt. A hálózat logikailag elkülönített részlegeket tartalmaz VLAN-nal, támogatja a dinamikus útvonalválasztást, és biztonságos, titkosított kapcsolatot biztosít VPN által távoli telephelyek között.

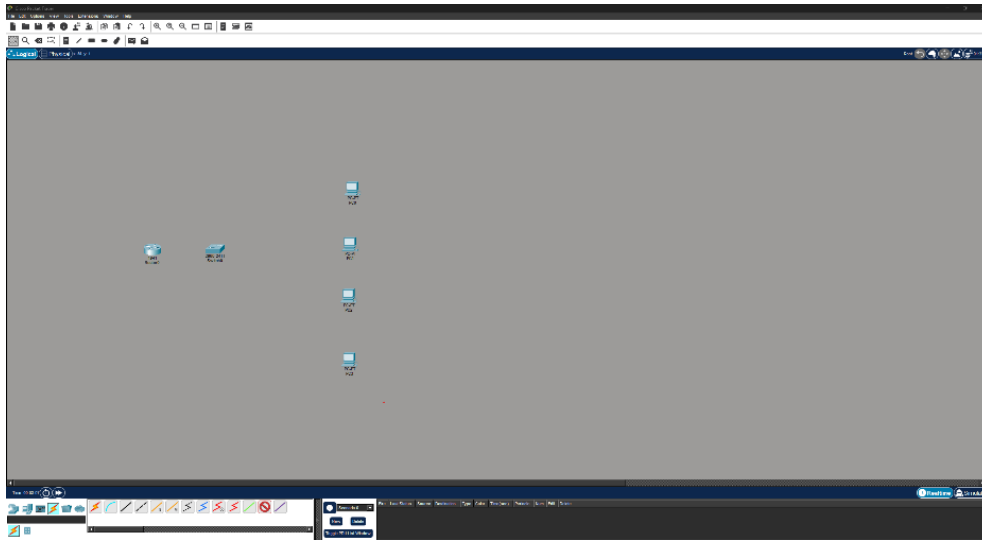
2. IP CÍMZÉSI TERV

A hálózat tervezése során VLSM (Variable Length Subnet Masking) technológiát használtam a hatékony címkiosztás eléréséhez.

Device / Interface	IP Address	Subnet Mask	VLAN / Purpose
Router 0 – G0/0.10	192.168.10.1	255.255.255.192 (/26)	VLAN 10 Default Gateway (IT Department)
Router 0 – G0/0.20	192.168.10.65	255.255.255.192 (/26)	VLAN 20 Default Gateway (HR Department)
Router 0 – S0/0/0	10.0.0.1	255.255.255.252 (/30)	WAN Point-to-Point Link
Switch 0 – VLAN 1	Assigned via Mgmt	-	Management Layer 2 Switch Interconnection

3. HÁLÓZATI TOPOLOGIA

A hálózat tervezése során a redundanciát és a szegmentációt tartottam fontos szempontnak. A topológia magába foglalja a központi routereket, a redundáns kapcsolatokkal ellátott switcheket és a végpontokat (1. ábra).



1. ábra A hálózat logikai topológiája Packet Tracerben

4. LAYER2 ÉS VLAN-OK

A broadcast tartományok csökkentése és biztonság érdekében VLAN szegmentációt alkalmaztam. A switchek közötti kapcsolaton LACP EthernetChannelt konfiguráltam a sávszélesség növelésére.

VLAN KONFIGURÁCIÓ (2. ÁBRA)

Az alábbi képen látható a VLAN adatbázis létrehozása és elnevezése a Switch CLI felületén .

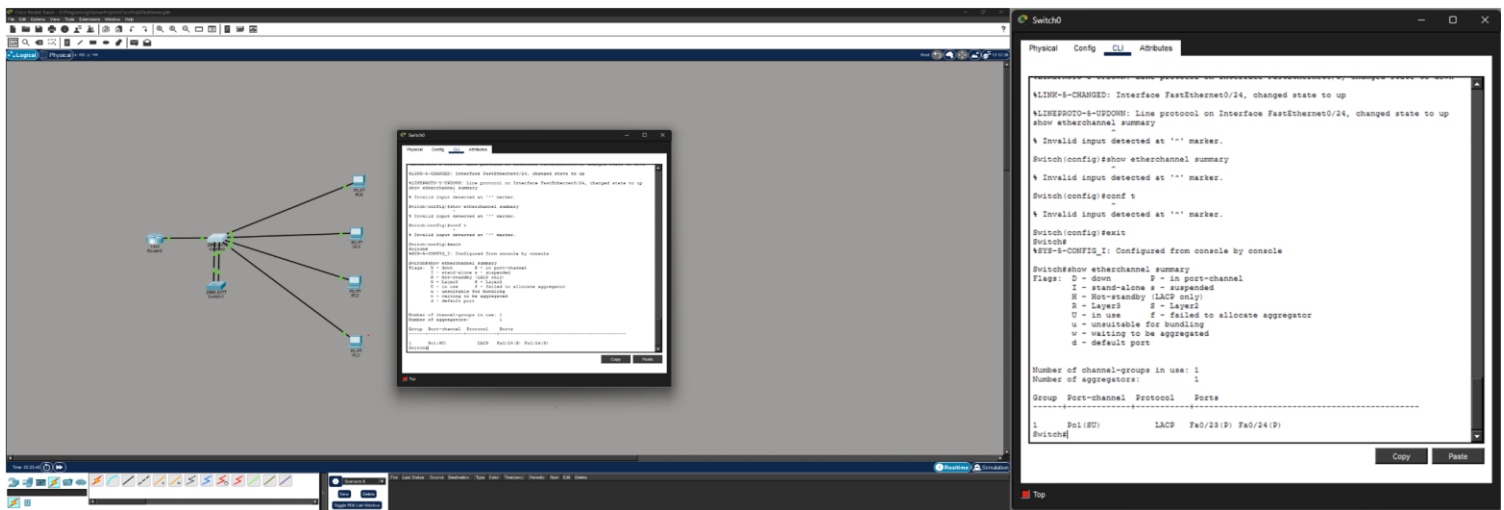
```
Switch0
Physical Config CLI Attributes
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up

Switch>enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT_Department
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR_Department
Switch(config-vlan)#exit
% Invalid input detected at '^' marker.
Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/1-10
Switch(config-if-range)#switchport mode access
% Invalid input detected at '^' marker.
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fastEthernet 0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

2. ábra VLAN létrehozása és elnevezése (IT és HR)

ETHERCHANNEL (3. ÁBRA)

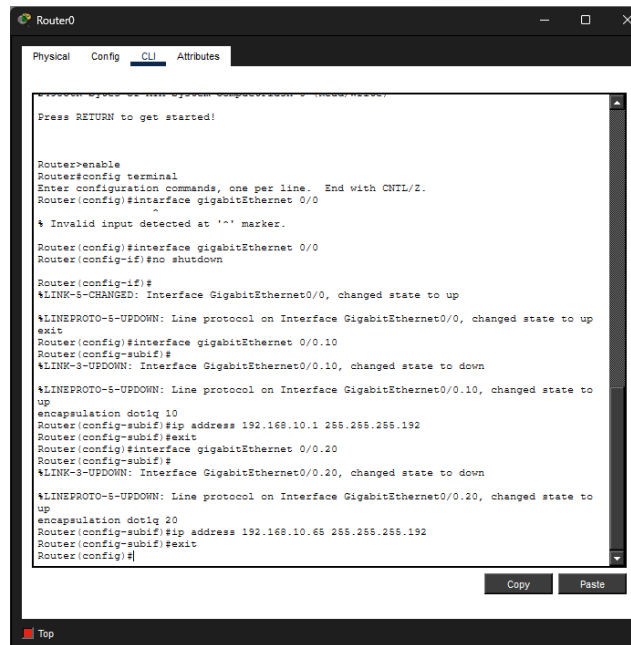
A show etherchannel summary parancs azt igazolja, hogy a port-aggregáció sikeresen létrejött.



3. ábra Az EtherChannel (PO1) státusz ellenőrzése

5. HÁLÓZATI RÉTEG ÉS ROUTING (LAYER 3)

A VLAN-ok közötti forgalomirányítást „Router on a Stick” módszerrel oldottam meg, 802.1Q enkapszulációt használva a subnetinterface-eken (4. ábra).

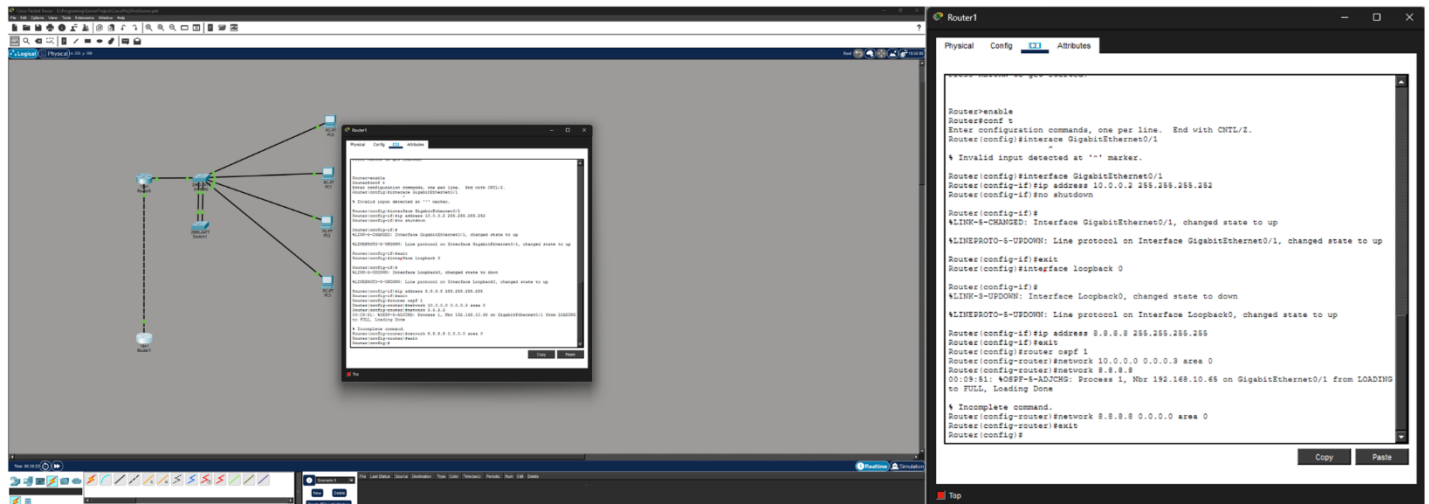
The screenshot shows the CLI of Router0 with the following configuration commands:

```
Router>enable
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.192
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1q 20
Router(config-subif)#ip address 192.168.10.65 255.255.255.192
Router(config-subif)#exit
Router(config)#
```

4. ábra Subinterface konfiguráció a routeren

OSPF DINAMIKUS ROUTING (5. ÁBRA)

A hálózat skálázhatósága érdekében OSPF protokolt konfiguráltam, amely automatikusan hirdeti a csatlakoztatott hálózatokat.

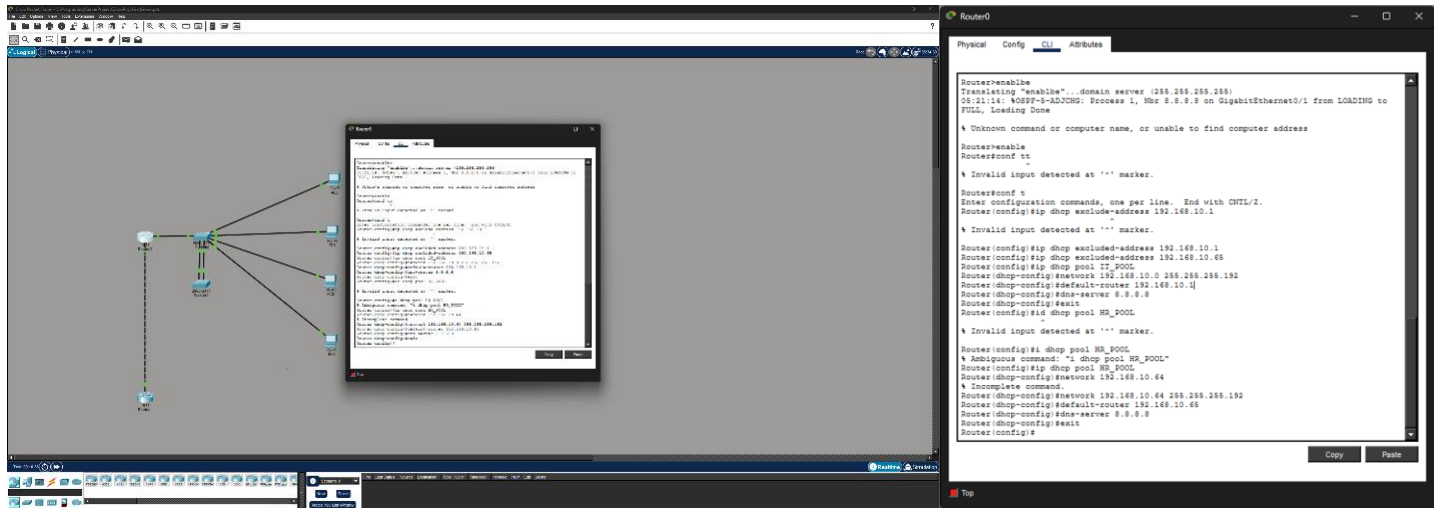


5. ábra OSPF hálózatok hirdetése

6. INFRASTRUKTÚRA SZOLGÁLTATÁSOK (DHCP)

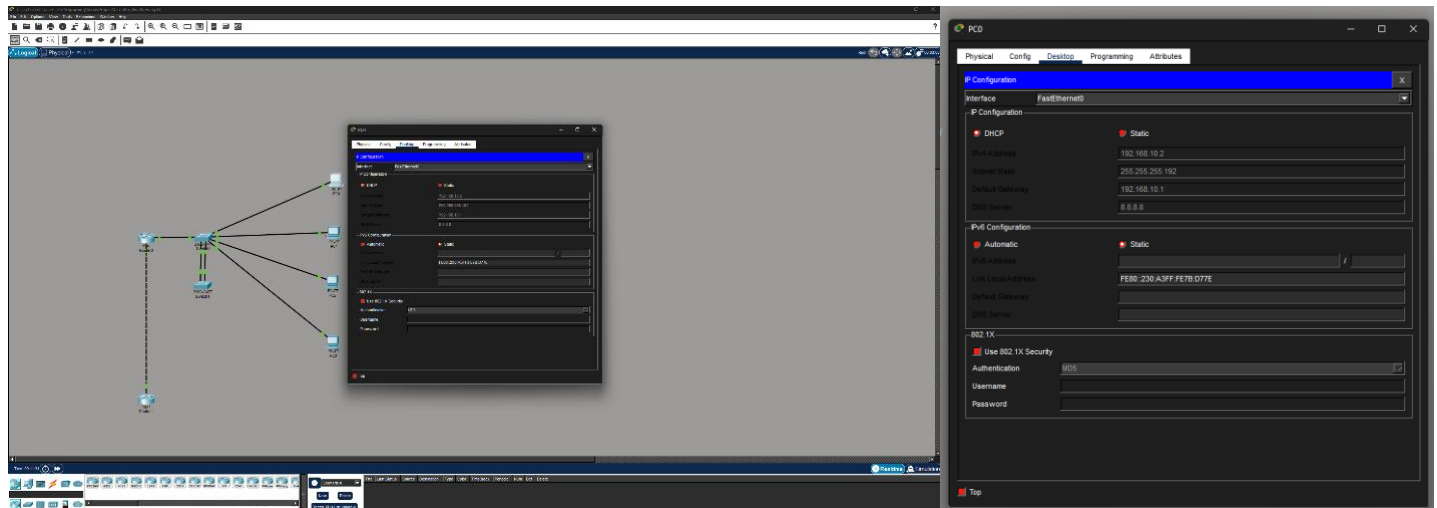
A végpont IP-cím kiosztását a routerre bízott DHCP szolgáltatás végzi.

SZERVER OLDALI BEÁLLÍTÁS: A ROUTEREN LÉTREHOZOTT POOL-OK ÉS A KIZÁRT CÍMEK BEÁLLÍTÁSA (6. ÁBRA).



6. ábra DHCP pool konfigurációja

KLIENS OLDALI ELLENŐRZÉS: A PC SIKERESEN MEGKAPTA AZ IP-CÍMET, AZ ÁTJÁRÓT ÉS A DNS SERVER CÍMÉT (7. ÁBRA).



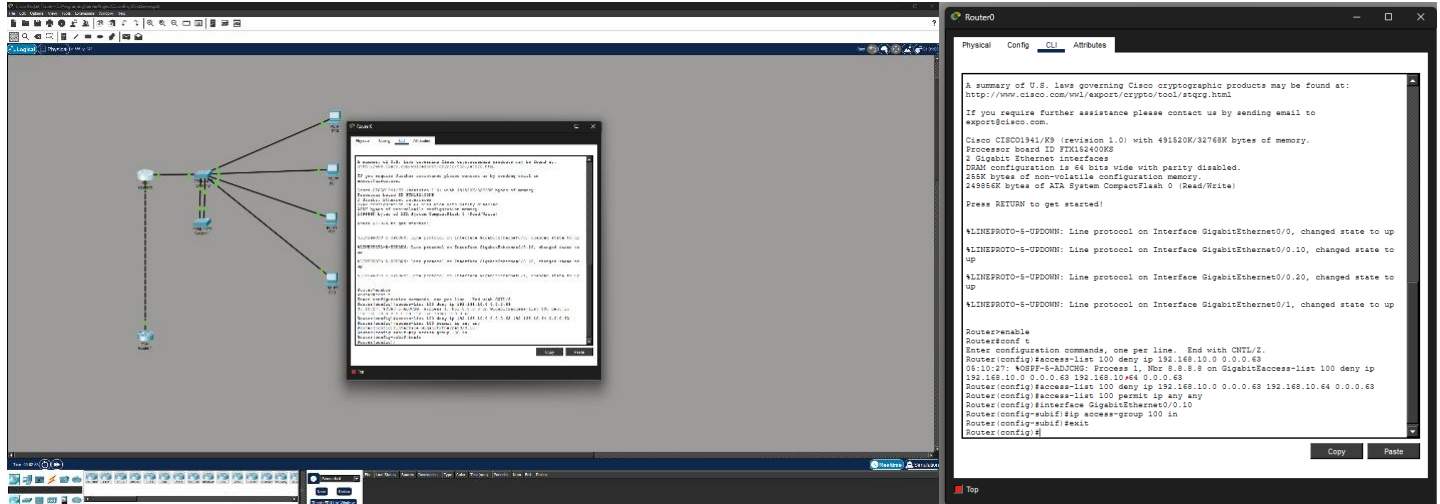
7. ábra Sikeres IP-cím kérés kliens oldalon

7. HÁLÓZATI BIZTONSÁG

Access Control List (ACL)

Biztonsági okokból szigorítottam a részlegek közötti kommunikációt. A 100 -as számú kiterjesztett ACL letiltja az IT hálózat hozzáférést a HR hálózathoz, de minden mást engedélyez.

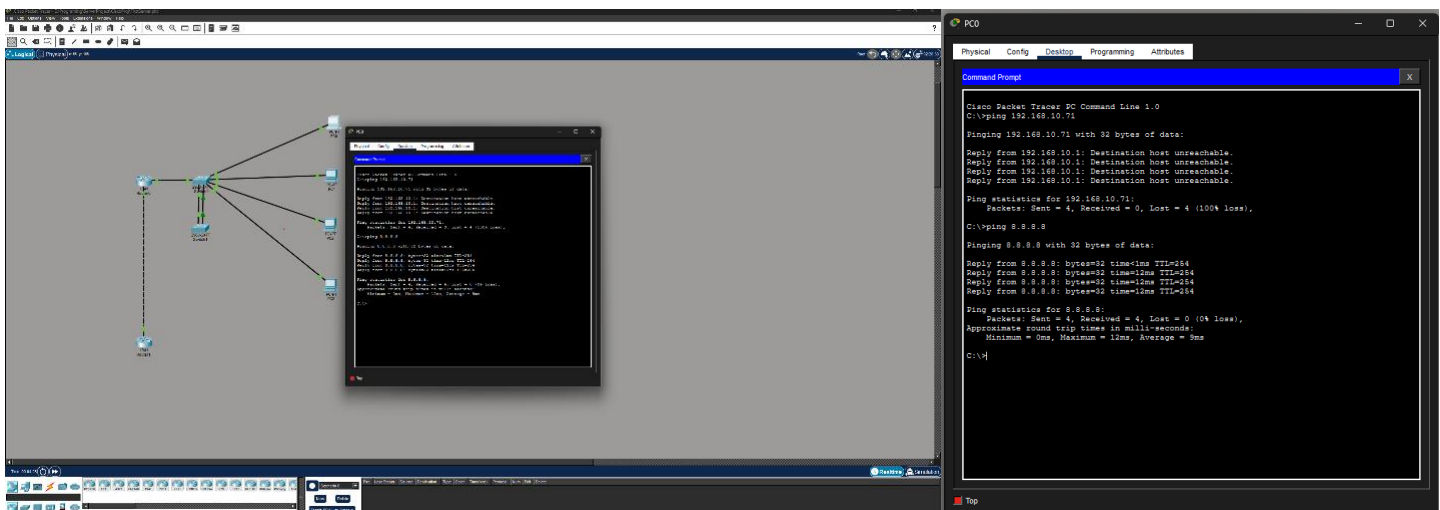
KONFIGURÁCIÓ (8. ÁBRA):



8. ábra Extended ACL szabályok felvétele

VERIFIKÁCIÓ:

A teszt során a ping parancs „Destination host unreachable” választ ad, ami igazolja, hogy az ACL sikeresen blokkolja a tiltott forgalmat (9. ábra).

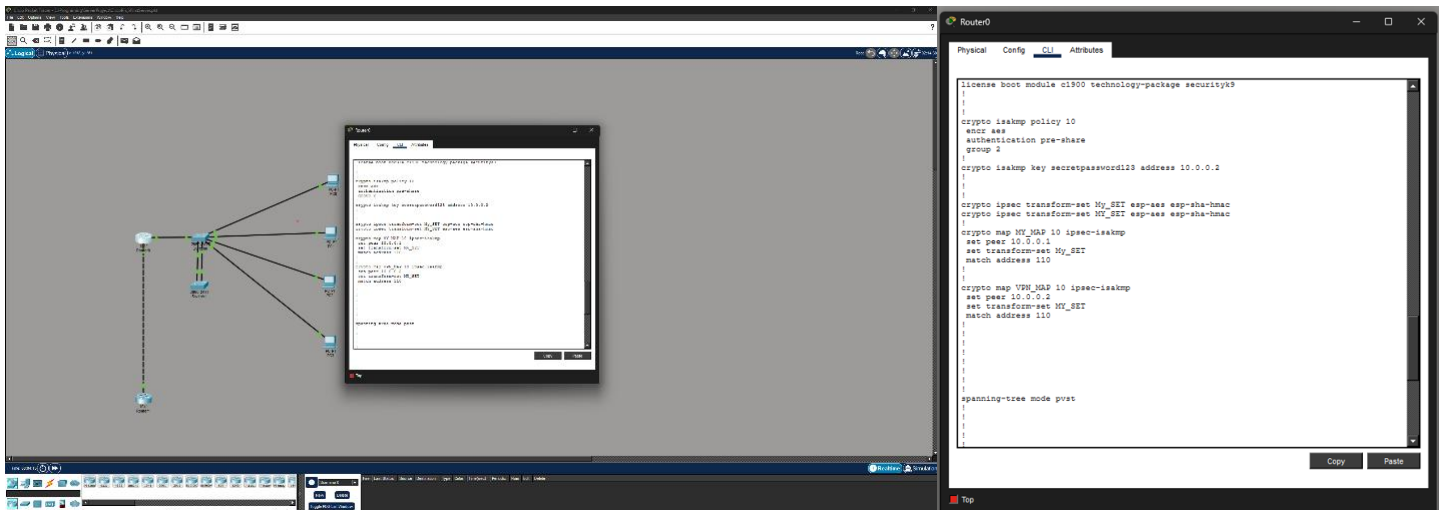


9. ábra ACL tesztelés

SITE TO SITE IPSEC VPN

A távoli telephelyek közötti biztonságos adatátvitelhez titkosított VPN alagutat építettem ki.

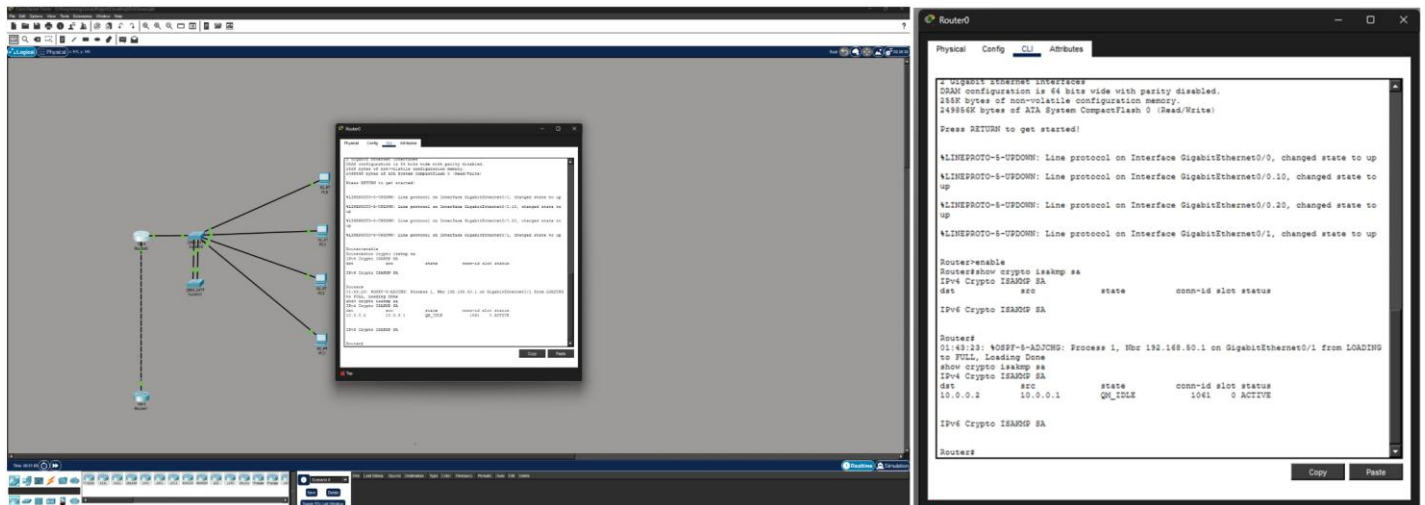
VPN konfiguráció (Crypto Map és Policy), (10. ábra)



10. ábra VPN policy és Transform Set beállítások

KAPCSOLAT STÁTUSZA:

A `show crypto isakmp sa` parancs kimenetében a `QM_IDLE` állapot jelzi, hogy a biztonságos csatorna aktív és működőképes (11. ábra).



11. ábra Aktív VPN alagút ellenőrzése

Small and Medium Enterprise Network Infrastructure Implementation Documentation

CISCO PACKET TRACER SIMULATION

Prepared by: Dávid Balogh
Date: January 13, 2026
Version: 1.0
Technologies: Cisco IOS, OSPF, VLAN, ACL, IPsec VPN, DHCP

1. INTRODUCTION

This document describes the design and configuration of a medium-sized enterprise network environment. The primary goal of the project was to establish a secure, scalable, and redundant infrastructure network capable of supporting modern corporate operations. The implemented architecture features logically isolated departments utilizing Virtual Local Area Networks (VLANs), implements robust dynamic routing protocols to guarantee high availability, and integrates a secure, encrypted connection via Site-to-Site VPN tunnels to safely link remote corporate offices over public networks.

2. IP ADDRESSING PLAN

During the network planning phase, Variable Length Subnet Masking (VLSM) technology was systematically applied to ensure efficient, optimized IP address allocation and to mitigate unnecessary address wasting across subnet segments.

Device / Interface	IP Address	Subnet Mask	VLAN / Purpose
Router 0 – G0/0.10	192.168.10.1	255.255.255.192 (/26)	VLAN 10 Default Gateway (IT Department)
Router 0 – G0/0.20	192.168.10.65	255.255.255.192 (/26)	VLAN 20 Default Gateway (HR Department)
Router 0 – S0/0/0	10.0.0.1	255.255.255.252 (/30)	WAN Point-to-Point Link
Switch 0 – VLAN 1	Assigned via Mgmt	-	Management Layer 2 Switch Interconnection

3. NETWORK TOPOLOGY

The layout was drafted with a strong emphasis on redundancy and broadcast traffic segmentation. The topology incorporates centralized core routers, switches provisioned with high-bandwidth redundant paths, and dedicated local endpoints.

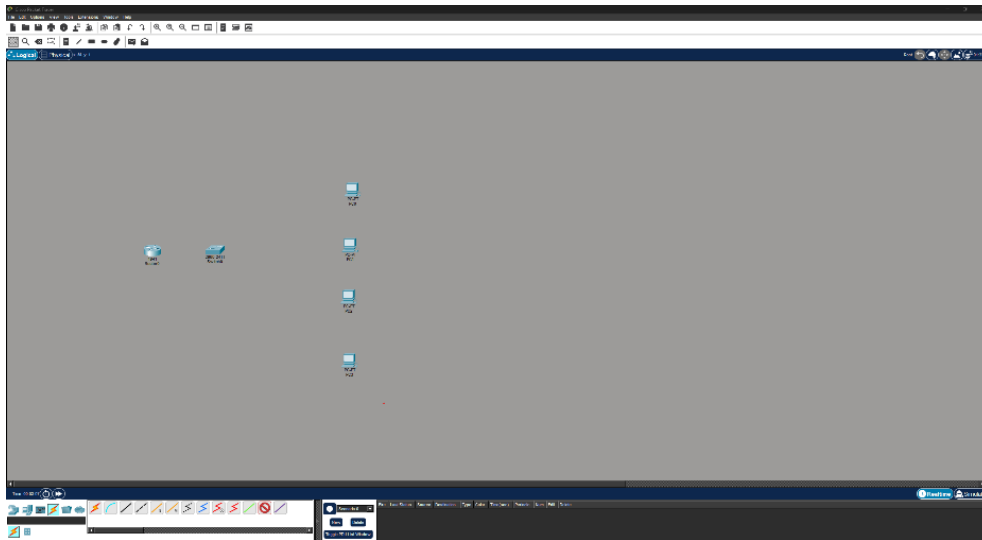


Figure 1: Logical network topology view inside Cisco Packet Tracer workspace.

4. LAYER 2 AND VLAN CONFIGURATION

To restrict broadcast domains and reinforce internal security boundaries, strict VLAN segmentation was applied. To maximize inter-switch link capacity and provide immediate hardware failover capabilities, a Link Aggregation Control Protocol (LACP) EtherChannel bundle was deployed.

VLAN SETUP

The corporate switch command line interface was utilized to declare and map out the corporate logical divisions as follows:

```
Switch0
Physical Config CLI Attributes
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up

Switch>enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT_Department
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR_Department
Switch(config-vlan)#exit
% Invalid input detected at '^' marker.

Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/1-10
Switch(config-if-range)#switchport mode access
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fastEthernet 0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

Figure 2: Creation and explicit naming of functional VLANs (IT_Department and HR_Department).

ETHERCHANNEL VERIFICATION

Execution of trunk aggregation diagnostic tools guarantees complete structural alignment:

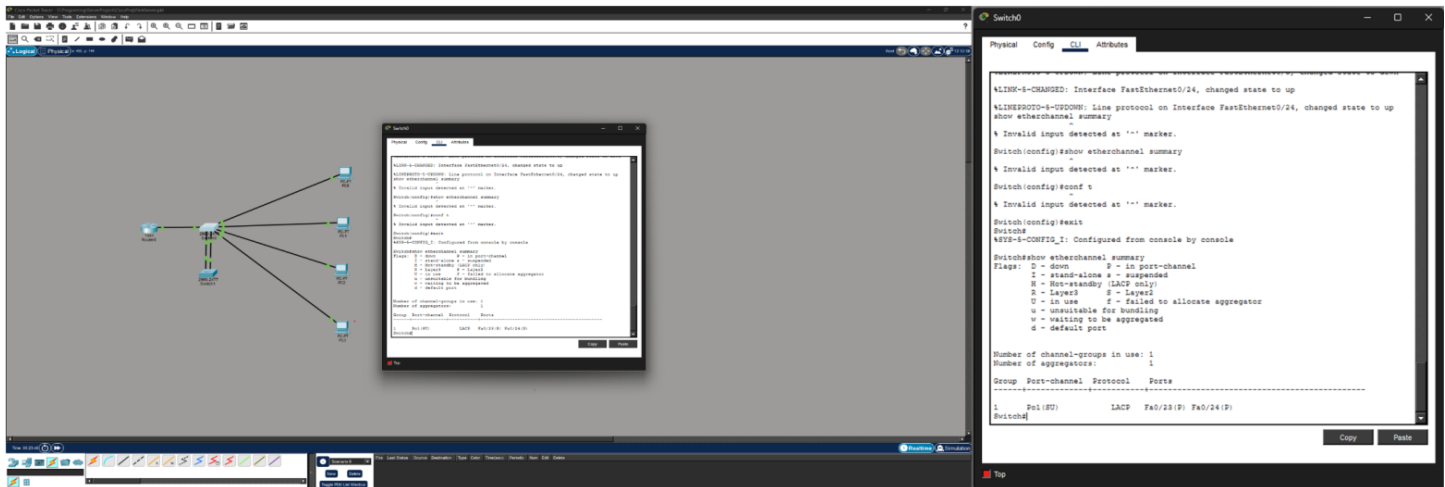


Figure 3: Verification of the EtherChannel (PO1) operational status using the show etherchannel summary command.

5. NETWORK LAYER AND ROUTING (LAYER 3)

Inter-VLAN routing services were implemented via a classic "Router-on-a-Stick" deployment strategy, binding multiple virtual subinterfaces to physical trunk connections through IEEE 802.1Q encapsulation standards.

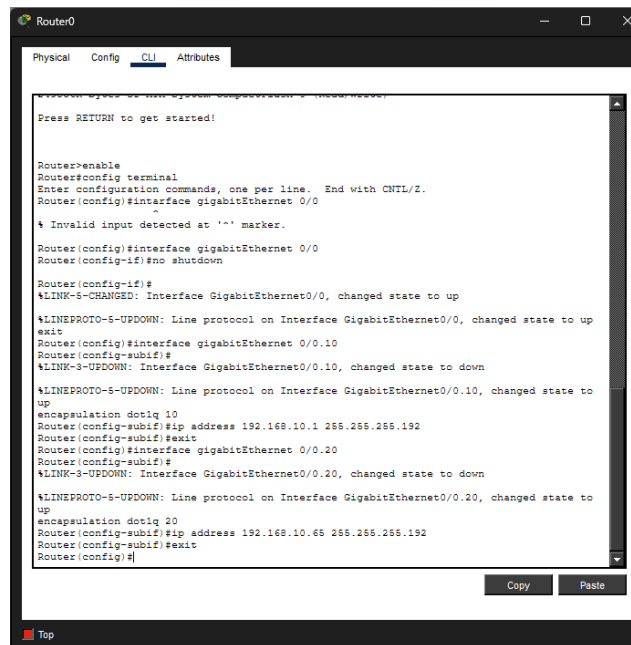


Figure 4: Subinterface logical separation and IP gateway binding configurations on Router 0.

OSPF DYNAMIC ROUTING

To promote seamless scalability and instantaneous topology convergence across the enterprise boundaries, Open Shortest Path First (OSPF) routing was initiated to dynamically discover and advertise active internal prefixes.

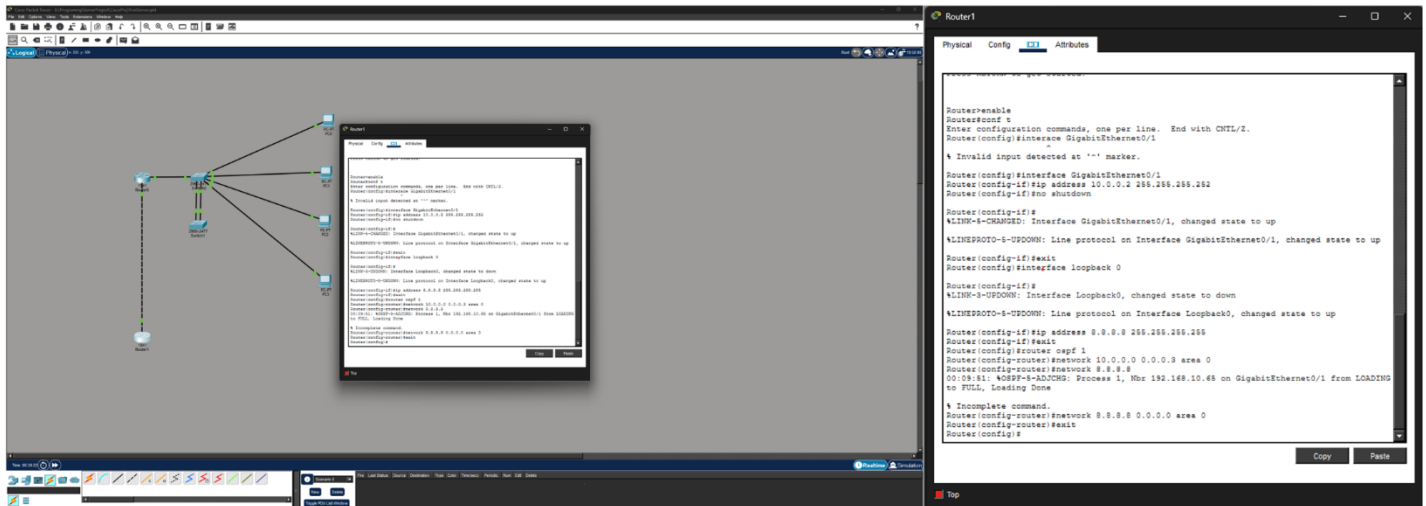


Figure 5: OSPF network statement publications and automated adjacency formations.

6. INFRASTRUCTURE SERVICES (DHCP)

Automated endpoint addressing services are completely offloaded to localized DHCP pools managed directly by the central routing platform.

- **Server-Side Architecture:** Implementation of explicit exclusion spaces protecting critical network resources alongside pool configurations.

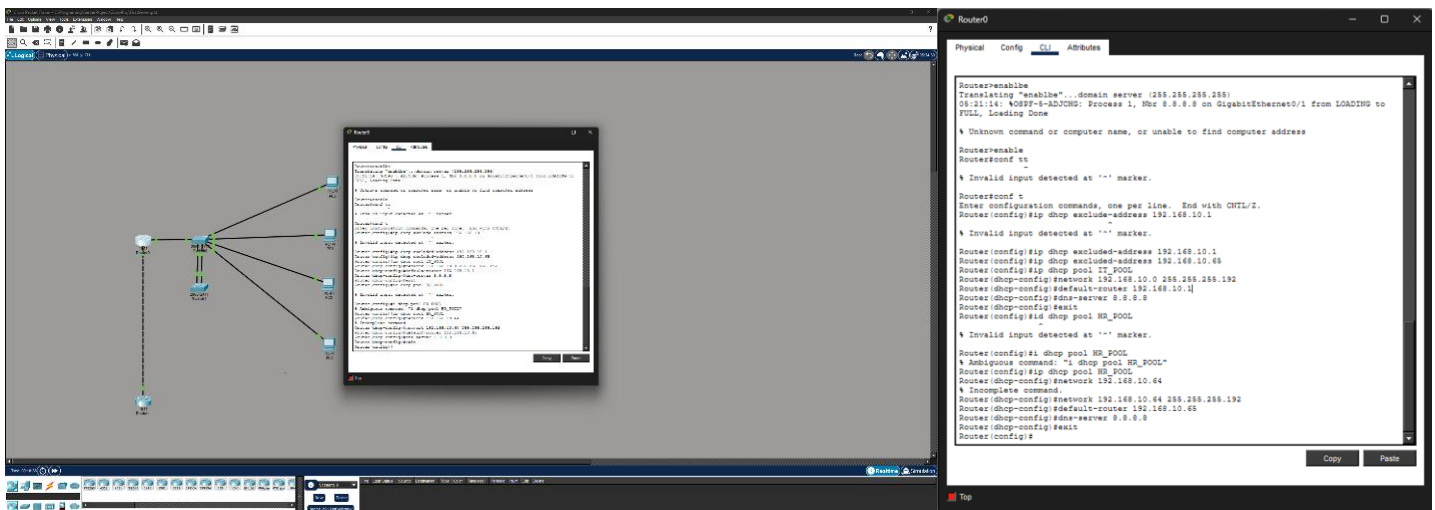


Figure 6: DHCP address pool exclusions, network boundaries, and lease parameters setup.

- **Client-Side Verification:** Connected terminal stations successfully lease configuration metrics dynamically.

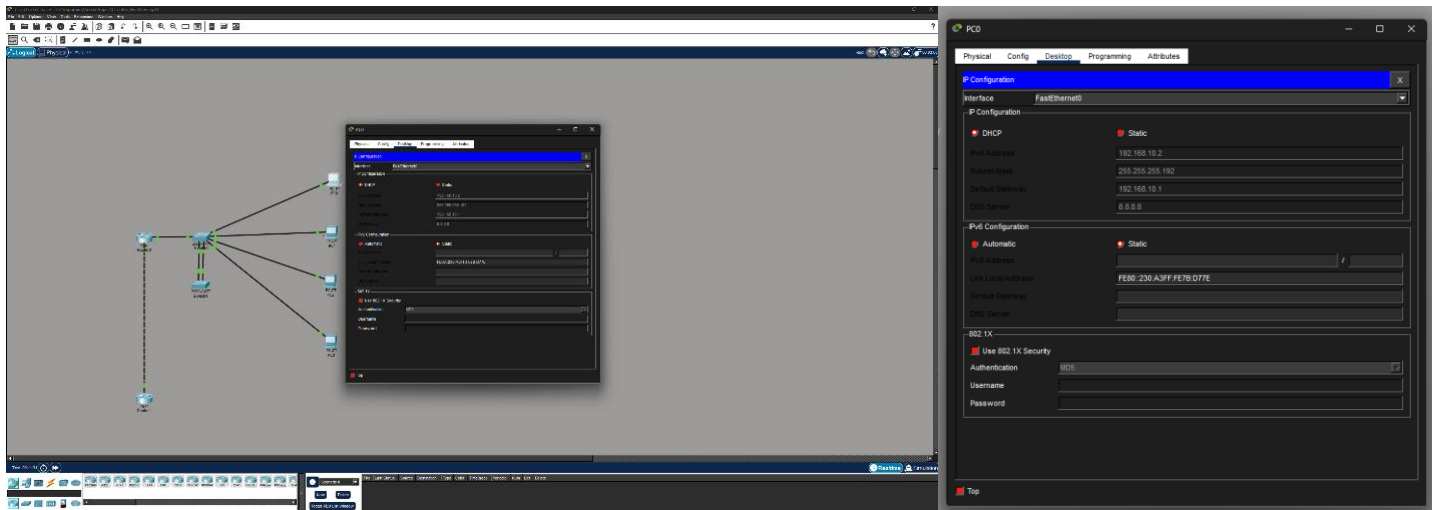


Figure 7: Target client workstation confirming valid IP parameters, default gateway, and corporate DNS records.

7. NETWORK SECURITY

ACCESS CONTROL LISTS (ACL)

To fulfill restrictive security blueprints, inter-departmental traffic flows were carefully regulated. An extended IP Access Control List (numbered 100) was introduced to completely isolate the HR environment from inbound IT scans while maintaining global internet access paths.

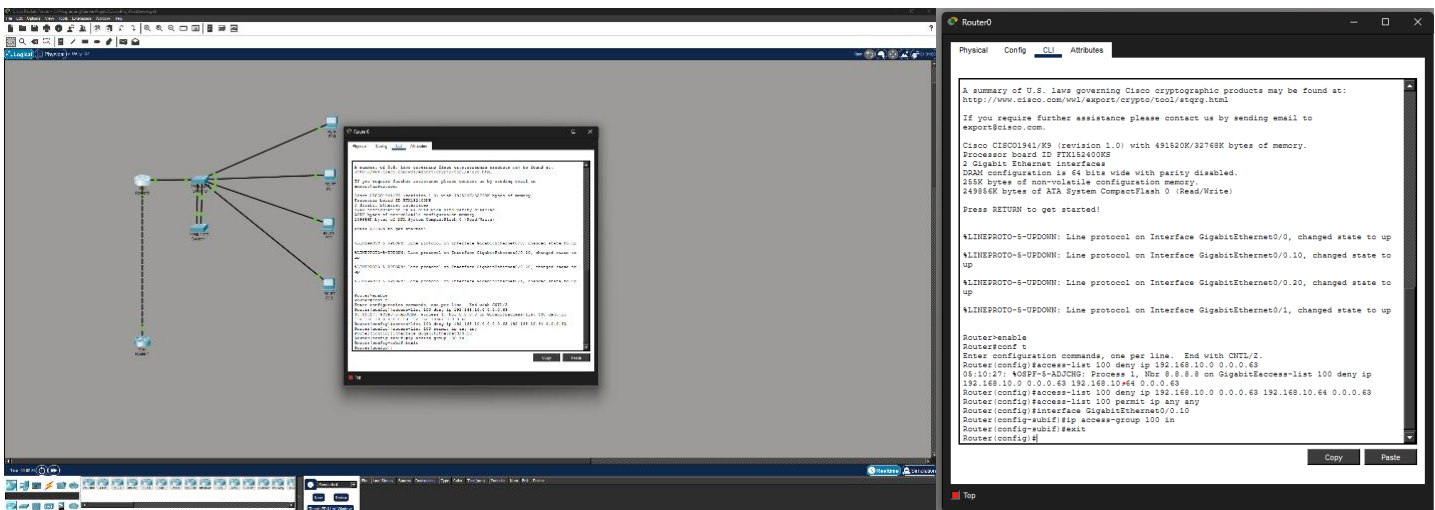


Figure 8: Definition and application of Extended ACL 100 inbound rules.

Verification: Active diagnostic ping attempts securely terminate with explicit ICMP notifications confirming proper filtering mechanics.

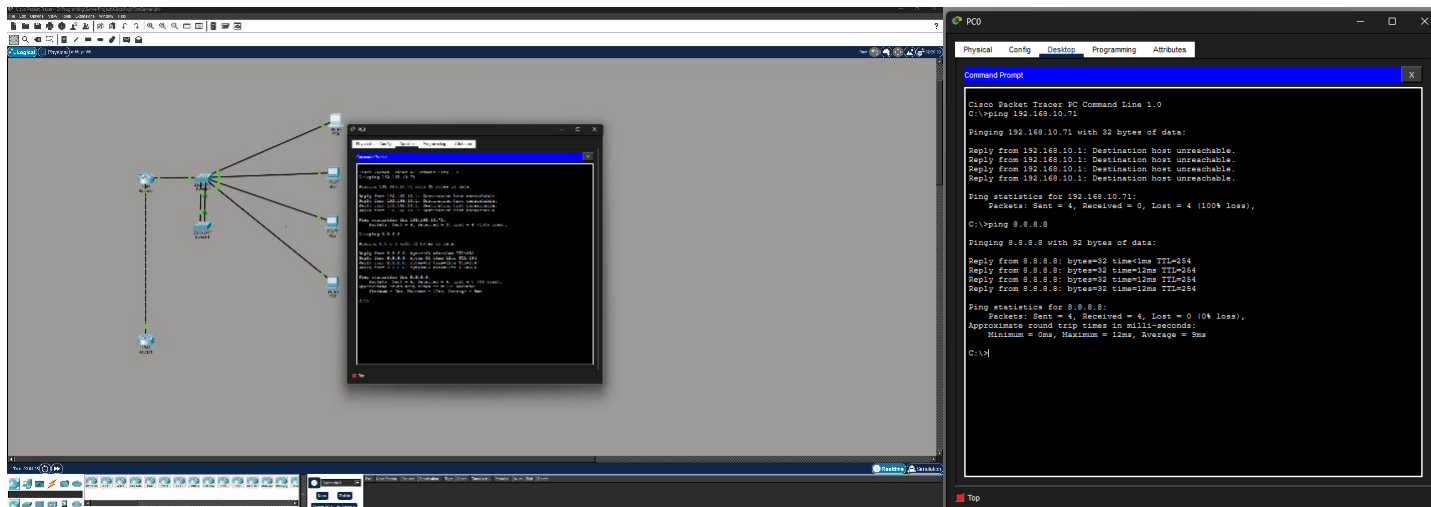


Figure 9: Firewall confirmation displaying "Destination host unreachable" status for blocked inter-VLAN routes.

SITE-TO-SITE IPSEC VPN

To guarantee absolute data confidentiality across public infrastructure paths separating external nodes, a cryptographically guarded IPsec VPN tunnel was provisioned using strong advanced encryption standards.

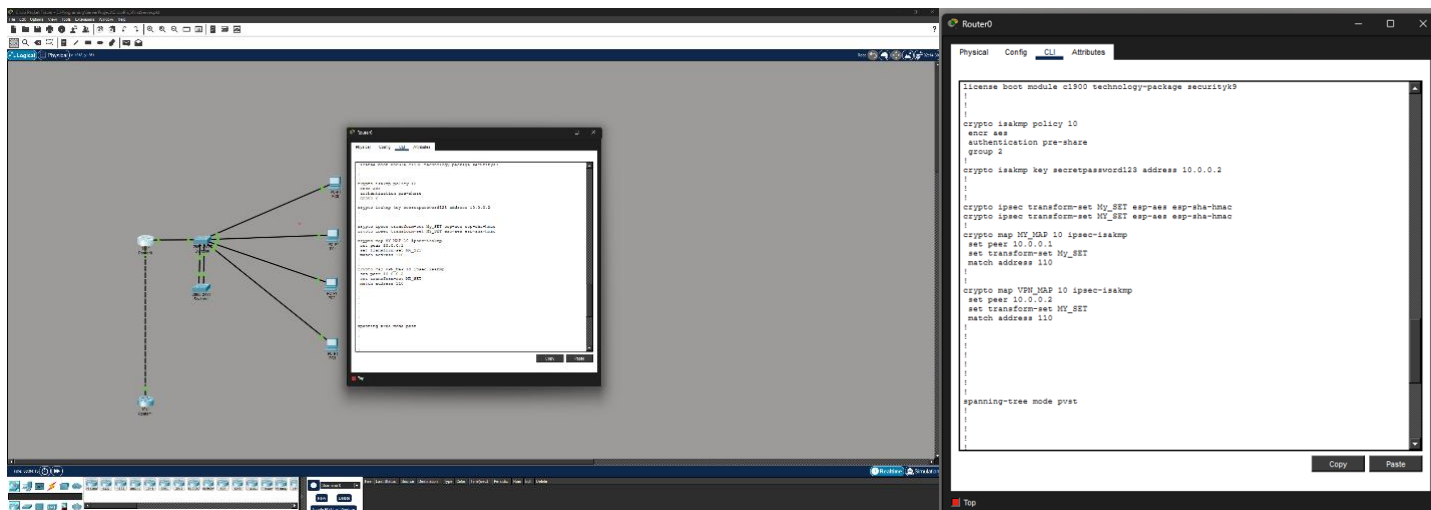


Figure 10: Crypto Isakmp Policy mappings, Pre-Shared Key strings, and Transform-Set rule declarations.

Tunnel Convergence Verification: Real-time security association validations prove functional state stability:

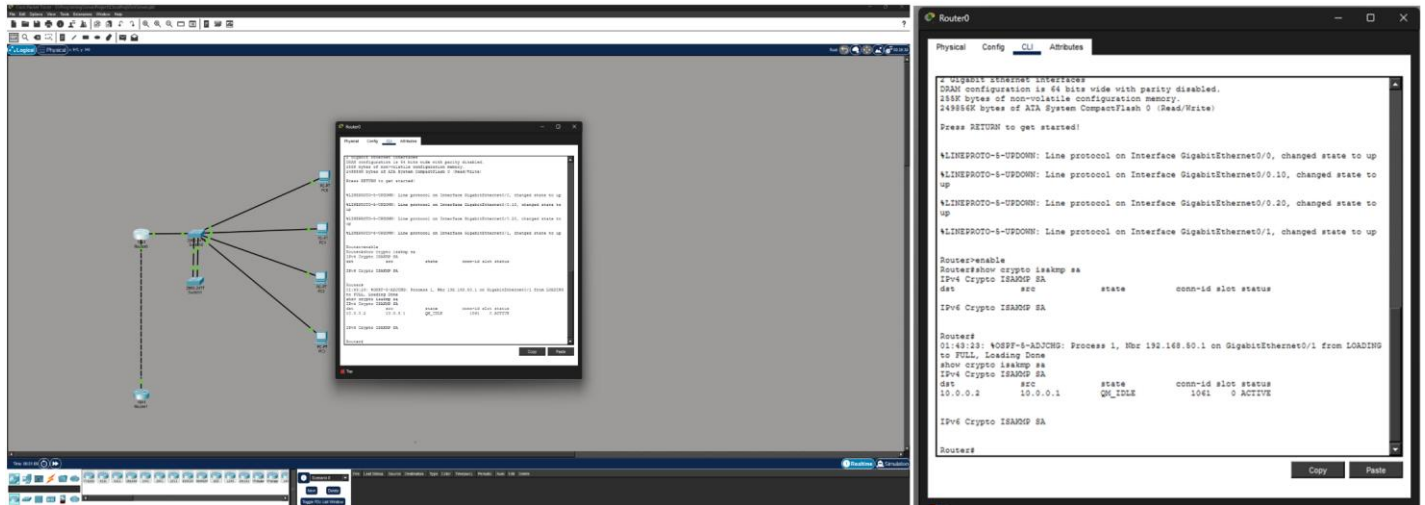


Figure 11: Active VPN channel validation indicating a successful QM_IDLE active operational state.